

Manual QA: Shmerling Play UI (Phase 1–4)

Product surface: Web `/play` and Desktop Electron Play shell (shared UI).

Scope: Phase 1–2 = single-player vs local Brain. Phase 3 = OnlineMode core play on `/play`. Phase 4 = draw, rematch (with color choice), reconnect countdown. Classic `/game` remains as fallback.

Last updated: 2026-08-01 (security remediations + Prefer-Play UCI)

Use this document as the living checklist. Add Phase 5+ sections at the bottom without rewriting earlier phases.

How to use this checklist

- 1. **Restart the app/server** before a full pass (`npm start` for web; restart Electron for desktop).
- 2. **Hard-refresh** the Play page (`Cmd+Shift+R` / `Ctrl+Shift+R`) so scripts and CSS reload.
- 3. For each test: follow **Steps**, then mark **Pass** only if every **Expect** item is true.
- 4. Record failures with: test id, role (Admin/Partner/Member), browser or Electron, and a short note.
- 5. Prefer a clean profile or known test users (`e2e_web_member`, `e2e_web_partner`, Admin) when roles matter.

Roles (quick reference)

Capability	Admin	Partner	Member
New Play UI (<code>/play</code>)	Yes	Yes	Yes
Position setup	Yes	Yes	Hidden
Brain config panel	Yes	Yes	Hidden
Saved games / review sidebar	Yes	Yes	Typically limited / hidden tools
Admin menu	Yes	No	No
Debug (classic <code>/game</code>)	Yes	Yes	No

Pre-flight

P0 — Environment ready

Steps

1. Confirm MongoDB is running (local or configured remote).
2. Start the web server (`npm start`) or open the desktop app.
3. Log in as the role you intend to test.
4. Open `/play` (web) or the desktop Play window.
5. Hard-refresh once.

Expect

- Play shell loads (board frame, action rail, header clocks area).
- No blank screen or console errors that block interaction.
- Idle status invites starting a new game (or similar).

P1 — Hard refresh after code changes

Steps

1. After pulling or receiving Play/session changes, restart the server.
2. Hard-refresh `/play`.

Expect

- New session scripts load (`/app/session/*.js`, `/app/play-ui/*.js`).
 - No stale cached `desktop-play.js`.
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Phase 1 — Play shell modules (feature parity)

Phase 1 carved presentation/policy into `src/play-ui/*` without changing product behavior. These tests confirm the shell still works as a whole.

1.1 — New game as White (Play Now / New game)

Steps

1. From Home, use **Play Now**, or on `/play` open **New game**.
2. Choose color **White**, engine **Brain 4.3** (or default), allow undo on, a comfortable thinking time (e.g. 5–10s).
3. Start the game.

Expect

- Board appears from White's view (White at bottom).
- Header shows your username vs the Brain label (not a generic "Player" if you are logged in).
- Clocks show; White's clock is the side to move / running (or ready to run).
- **Resign** enabled on your turn; **Draw** disabled vs engine.
- Status indicates it is your move (or equivalent).

1.2 — New game as Black

Steps

1. Start a new game as **Black**.
2. Wait for the engine's first move.

Expect

- Board flipped (Black at bottom) if that is the product default for playing Black.
- Engine moves first with animation.
- After the engine move, it is your turn; clocks switch appropriately.
- Header names still correct (you as Black, Brain as White).

1.3 — Engine selection persists (Brain 4.2 / 4.3)

Steps

1. Start a game with **Brain 4.2**.
2. Finish or leave.
3. Open New game again (or Play Now).

Expect

- Engine choice remains **Brain 4.2** (or the last explicit choice), not silently reset to an old default without reason.
- Starting again with **Brain 4.3** then sticks on the next open.

1.4 — Drag move (mouse drag preference)

Steps

1. Preferences → Mouse control → **Drag**.
2. Start a game as White.
3. Drag a pawn e2→e4.

Expect

- Move is applied on the board.
- Moves panel shows the ply.
- Clocks switch after your move.
- Engine replies (animation + move in the list).

1.5 — Double-click / click-to-move preference

Steps

1. Preferences → Mouse control → **Double-click** (or click-to-move mode as labeled).
2. Start a game as White.
3. Select a piece, then the target square, to play e2–e4.

Expect

- Legal targets highlight when configured.
- Move records the same as drag mode.
- Illegal clicks do not corrupt the board.

1.6 — Show available moves toggle

Steps

1. Preferences → enable **Show available moves**.
2. In a game, select a piece (click-to-move) or begin a drag as applicable.
3. Disable the preference and try again.

Expect

- With it on: legal destination hints appear.
- With it off: hints do not appear (or match product rule).

1.7 — Thinking time preference

Steps

1. Preferences → set thinking time to a low value (e.g. 2s).
2. Play a move and wait for the engine.
3. Increase to a higher value and play another game.

Expect

- Engine search duration roughly tracks the preference (not stuck on a hard-coded old value).

- Preference survives refresh.

1.8 — Clocks count down for side to move

Steps

1. Start a game with a long game clock (e.g. 90 minutes) if configurable.
2. On your turn, watch the active clock for several seconds.
3. Make a move; watch the opponent/engine clock while the engine thinks / after turn switch.

Expect

- Only the side to move's clock decreases.
- After a move, the other side's clock becomes active.
- Clocks stop when the game ends.

1.9 — Flip board

Steps

1. During an active game, click **Flip**.
2. Flip again.

Expect

- Board orientation reverses (files/ranks and piece placement).
- You can still move your pieces on your turn.
- Flip does not change whose turn it is.

1.10 — Last move arrow

Steps

1. Play at least one move (and optionally wait for the engine reply).
2. Click **Last move**.
3. Click again to toggle off if that is the behavior.

Expect

- An arrow (or last-move highlight) appears for the last ply.
- Toggle clears it without breaking input.

1.11 — Moves panel updates

Steps

1. Play several plies (you + engine).
2. Scroll the moves list if needed.

Expect

- Moves appear in order with sane notation.
- List stays in sync after undo/redo (see Phase 2 tests).
- No duplicate rows for a single ply.

1.12 — Action rail enable/disable (idle vs active)

Steps

1. On idle Play (no active game): observe Resign / Undo / Redo / Draw / Flip.
2. Start a game: observe again on your turn, while the engine thinks, and after game over.

Expect

- Idle: play actions locked as designed; New game available.
- Active your turn: Resign on; Draw off vs engine; Undo/Redo follow allow-undo and history.
- While engine thinks / animating: destructive actions disabled.
- After game over: Resign off; New game still usable.

1.13 — Preferences / themes / piece set (smoke)

Steps

1. Open Preferences from the Play chrome / account menu.
2. Change piece set or theme if available.
3. Close preferences and confirm the board still plays.

Expect

- UI updates without breaking Play.
- Game can still be started and moves played.

1.14 — Keyboard shortcuts (smoke)

Steps

1. On /play, try documented shortcuts (e.g. F2 for new-game related flow if enabled; Cmd/Ctrl+E evaluation if Partner/Admin).
2. Focus a text field (rename saved game if available) and press the same keys.

Expect

- Shortcuts work when not typing in an input.
- Shortcuts are ignored (or safely no-op) while typing in inputs.

1.15 — Evaluation display (Partner/Admin)

Steps

1. As Partner or Admin, start or load a position.
2. Trigger evaluation (Cmd/Ctrl+E or UI control).
3. Click away / dismiss if applicable.

Expect

- Evaluation appears in the status/UI without freezing the board.
- Dismiss clears overlay/title as designed.

1.16 — Setup vs Config mutual exclusion (Partner/Admin)

Steps

1. Open **Position setup**.
2. Observe the **Config** control.
3. Exit setup; open **Config**.
4. Observe **Position setup**.

Expect

- Only one of Setup/Config is active at a time.
- While one is open, the other control is disabled (cannot click-switch without exiting).
- Exiting restores normal idle/play chrome.

1.17 — Member cannot see Setup / Config

Steps

1. Log in as **Member**.
2. Open /play.

Expect

- Position setup and Brain config entry points are hidden or unavailable.
 - New game vs Brain still works.
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Phase 1 — Position setup (Partner/Admin)

1.18 — Enter position setup from menu

Steps

1. From idle Play, open **Position setup**.
2. Clear / place pieces using the setup UI (drag on board / palette as designed).

Expect

- Setup mode chrome appears.
- Human play on a live game is off.
- Status indicates setup (not a finished game).

1.19 — Setup draw status clears when material becomes sufficient

Steps

1. In setup, leave only two kings (insufficient material).
2. Confirm status shows a draw / insufficient material message and kings may show draw highlight.
3. Add a queen (or other piece) so material is sufficient.

Expect

- Draw highlights clear from the board.
- **Status line clears** the draw text (returns to setup guidance), not a stuck "Draw — Insufficient material".

1.20 — Validate and play from setup

Steps

1. Set a legal middlegame-like position (both kings, sensible pieces).
2. Set side to move / human color via the game-run options if shown.
3. Start **Play** from the position.

Expect

- Invalid positions are rejected with a clear alert (not a silent failure).
- Valid positions start an active game with clocks and correct turn.
- Engine moves if it is the AI's turn.

1.21 — Setup and Config do not both stay open

Steps

1. Enter setup, then attempt to open config (or the reverse).

Expect

- Mutual exclusion holds (see 1.16).
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Phase 1 — Brain config (Partner/Admin)

1.22 — Open config, change a value, play

Steps

1. Open **Config** (brain config panel).
2. Change a visible non-destructive setting if available; save/apply if required.
3. Close config; start a short game.

Expect

- Panel opens and closes cleanly.
 - Play still works afterward.
 - No leftover dialog lock (dialog0n) that blocks moves.
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Phase 1 — Saved games & review (Partner/Admin)

1.23 — Auto-save / manual save after a game

Steps

1. Play a few moves; resign or finish.
2. Check the saved games / positions list (Games sidebar).

Expect

- A bookmark appears (auto and/or after Save), with a sensible name.
- List can expand to show details.

1.24 — Load saved game into review

Steps

1. Click a saved game with moves to load it.
2. Confirm review navigation appears (Start / Back / Forward / End / Play).

Expect

- Board shows the final (or loaded) position.
- Review nav is visible when moves exist.
- Human play is disabled in review.

1.25 — Review Start / Forward / Back / End

Steps

1. With a multi-move saved game loaded, click **Start** (ply 0).
2. Click **Forward** several times.
3. Click **Back**.
4. Click **End**.

Expect

- Board and moves selection track the ply.
- No engine search starts during review navigation.
- Playback Play/Pause advances without crashing; Stop works.

1.26 — Review move-list click to ply

Steps

1. In review, click a move in the moves panel.

Expect

- Board jumps to that ply.
- Nav buttons update enabled/disabled state.

1.27 — Load from start (double-click behavior if present)

Steps

1. Double-click a saved game (or use the documented "load from start" affordance).

Expect

- Review opens at the start position when that feature exists.
- Forward still replays correctly.

1.28 — Edit saved position → setup

Steps

1. Use **Edit** on a saved position/game (Partner/Admin).
2. Confirm you enter position setup with that board.
3. Cancel or save as product allows.

Expect

- Review mode exits.
- Setup opens with the expected pieces.
- Leaving edit does not leave a half-active game session.

1.29 — Delete saved game

Steps

1. Delete one bookmark (and multi-select delete if available).
2. Confirm list refresh.

Expect

- Item removed.
- Status confirms deletion.
- Board is not left in a broken review state.

1.30 — Exit review into new game

Steps

1. Load a review.
2. Start **New game**.

Expect

- Review chrome clears.
 - Fresh game starts; engine and clocks work.
 - No stuck review nav bar.
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Phase 2 — GameSession / LocalEngineMode (ports-clean active play)

Phase 2 routes active single-player through GameSession + LocalEngineMode (commands/events). Tests below stress that path.

2.1 — Human move → session → engine reply

Steps

1. New game as White.
2. Play e2–e4.
3. Wait for the engine.

Expect

- Your move appears immediately (board + moves list).
- After paint, clocks switch; engine thinks; reply animates.
- No double status flashes for the same event.
- No second illegal “ghost” engine move.

2.2 — Username in header (not “Player”)

Steps

1. Log in with a real username.
2. Start a game as White and as Black.

Expect

- Header uses your username on your side.
- Brain label on the engine side.
- Refresh mid-game still shows the username (see 2.10).

2.3 — Draw button disabled vs engine (capabilities)

Steps

1. During an active engine game on your turn, inspect **Draw**.
2. If it is somehow clickable, click it.

Expect

- Draw is disabled for local engine mode.

- If clicked via forced enable, user is told draw offers are unavailable vs engine (legacy alert path).

2.4 — Resign (human)

Steps

1. Play at least one move.
2. Click **Resign** and confirm.

Expect

- Status: game over / you resigned (wording may vary).
- **Resigning king**: red square background and **90° rotation**.
- Clocks stop; Resign disables; moves list may show result.
- No duplicate conflicting status lines.

2.5 — Leave / Home mid-game resigns after you have moved

Steps

1. Play at least one move as White (or two plies if needed for Black).
2. Click Home / Leave.
3. Confirm the resign dialog if shown.

Expect

- Dialog explains the game will be resigned when applicable.
- After confirm, game is resigned and you leave Play (or return home).
- Leaving with no human moves may skip resign (product rule).

2.6 — Undo / Redo pair

Steps

1. New game with **allow undo** on.
2. Play a move; wait for engine reply (full pair).
3. **Undo**.
4. **Redo**.

Expect

- Undo removes your move and the engine reply (pair).
- Board, moves list, and clocks match the restored position.
- Redo restores the pair once; further redo stays disabled until another undo.

- Undo disabled while engine is thinking.

2.7 — Promotion (human)

Steps

1. Set up or play to a position where you can promote (setup→play is fine).
2. Move the pawn to the 8th rank.
3. Choose a piece in the promotion dialog (e.g. Queen).

Expect

- Promotion chooser appears only for you (not for the engine side).
- After choice, board shows the piece; moves list updates.
- Engine replies if it is the AI turn.
- No stuck "Choose promotion piece" status.

2.8 — Check indication

Steps

1. Play (or set up) a position that delivers check without mate.
2. Observe status and king highlight.

Expect

- Brief check status (or equivalent).
- Checked king highlighted (gold/check style — not resign red).
- Game continues.

2.9 — Checkmate / draw terminal outcomes

Steps

1. Deliver checkmate (setup a mate-in-one and play it), **or** reach a clear draw (stalemate / insufficient material from play/setup→play).
2. Observe status, highlights, clocks, and action rail.

Expect

- **One** clear terminal status (not double stacked messages).
- Checkmate: mated king red/checkmate style; clocks stop.
- Draw: draw status + draw king highlights as designed.
- Resign disabled; New game available.
- No engine search after the game is over.

2.10 — Refresh resumes in-progress game (web)

Steps

1. Start a game; play a few moves.
2. Hard-refresh /play (or reload).

Expect

- Position, moves, clocks, and turn restore.
- Username still correct.
- You can continue; engine still replies on its turn.

2.11 — Play from custom position (session attached)

Steps

1. Position setup → legal position with **engine to move** (human Black or White to move for engine).
2. Start play.
3. If engine to move, wait; if you to move, play one move.

Expect

- Session is active (undo/resign behave; engine runs when appropriate).
- No "stuck idle" where moves don't trigger the brain.

2.12 — Immediate resign (Brain resigns when lost)

Preconditions: Preferences → **Immediate resign** enabled; confirm it stays checked after refresh.

Steps

1. Enable Immediate resign; hard-refresh; confirm still checked.
2. Reach a position where the Brain is clearly lost / forced mate (use setup→play if needed so it is the engine's turn in a lost position), **or** play until the engine detects forced loss.
3. Wait for the engine's turn to resolve.

Expect

- Engine **resigns** instead of playing on (when detection fires).
- Status line reports resignation / game over.
- **Visual:** resigning king has **red background** and **90° tilt** (not only the status text).
- Clocks stop; no further engine moves.

2.13 — Immediate resign off

Steps

1. Disable Immediate resign.
2. Use a similar lost position for the engine.

Expect

- Engine still plays a move (may be a losing move) rather than auto-resigning, unless another rule ends the game.

2.14 — Browser context menu suppressed on Play

Steps

1. On `/play`, right-click the board, empty chrome, and header.
2. Right-click a saved game (Partner/Admin) if the custom menu exists.

Expect

- Native browser menu (Back / Reload / Ask Gemini / etc.) does **not** appear on Play.
- Custom app menus (e.g. saved-game context menu) still work when provided.

2.15 — Review mode attach/detach does not break next game

Steps

1. Load a saved game into review (Partner/Admin).
2. Navigate plies.
3. Start a new engine game.

Expect

- Review mode ends cleanly.
- New game engine turns work (session LocalEngineMode re-attached).
- No "engine never moves" regression.

2.16 — Out-of-time (if easy to reproduce)

Steps

1. If you can set a very short game time, start a game and let a side flag.
2. Otherwise skip and note "not run".

Expect

- Timeout status; clocks stopped; game over.
- Single timeout message (session-driven).

Cross-cutting regression matrix

Run these after any large Play/session change:

Area	Quick probe
New game W/B	1.1, 1.2
Move + engine	2.1
Undo/redo	2.6
Resign visual	2.4, 2.12
Refresh resume	2.10
Setup → play	1.20, 2.11
Review → new game	1.25, 2.15
Member restrictions	1.17
Context menu	2.14
Online invite / moves	3.1–3.4
Online resign / cancel	3.5–3.6
Online refresh / classic	3.7, 3.9

Phase 3 — OnlineMode on /play

Prerequisites: Two logged-in users (e.g. Member A and Member B) in two browsers or profiles. Friends of each other. Server restarted; hard-refresh /play.

Out of scope for Phase 3 (expect deferred / disabled): draw offers, rematch, chat, watchers on /play, reconnect forfeit countdown UI.

3.1 — Invite lands both players on /play

Steps

1. As user A, open Friends and invite user B to a game.

2. As user B, accept the invite.
3. As user A, wait for the accepted / establishing flow to finish.

Expect

- Both browsers navigate to a URL under `/play?id=...` (not classic `/game?id=`).
- Play shell loads (board, clocks, action rail, player names).
- User A is White; user B is Black (board orientation matches).

3.2 — Waiting for opponent (White)

Steps

1. As user A, send an invite and stay on the game page before B accepts (or use a flow where White is alone briefly).
2. Observe status / black player name while waiting.
3. Have B accept and join.

Expect

- While waiting: status like “Waiting for opponent...”; black name shows looking-for-opponent (or empty) until join.
- When B joins: black name updates to B’s username; status clears to a normal play hint.
- Resign is available once the opponent is present (or per product rules after join).

3.3 — Moves both directions

Steps

1. With both players connected on `/play?id=...`, as White play `e2–e4` (or any legal first move).
2. As Black, confirm the move appears animated and play a reply.
3. Continue for at least 3–4 half-moves each.

Expect

- Each side sees the opponent’s move animate onto the board.
- Moves table updates for both players.
- No “double move” / echo (your own move does not fire a second time from the network).
- Illegal moves are rejected locally (piece does not stick on an illegal square).

3.4 — Server clocks

Steps

1. During an online game on `/play`, watch both clocks for a full move cycle (White moves, Black moves).
2. Optionally refresh mid-game (see 3.7) and compare clock values look sane (not reset to full unless game is new).

Expect

- Side to move's clock counts down.
- After a move, clocks switch sides.
- After the opponent's move, your remaining time looks consistent with play (server-driven timers / `moveTime`).

3.5 — Cancel before the first move

Steps

1. Start a fresh online game (invite accepted, **0 moves** played).
2. As either player, use **Resign** or **Home** leave while still at move 0.
3. Observe the other player (if still connected).

Expect

- Game cancels (status "Game cancelled" or equivalent).
- No resign result / red king required for a zero-move cancel.
- Opponent is notified or lands in a cancelled / ended state (not stuck waiting forever).

3.6 — Resign after moves

Steps

1. Play at least one move in an online `/play` game.
2. As one player, resign (confirm dialog if shown).
3. Observe both clients.

Expect

- Game ends; status indicates resign / winner.
- Resigned king shows red square + 90° tilt (same visual as SP resign).
- Opponent sees "Opponent resigned" (or equivalent) and the same terminal board.
- Draw / rematch remain unavailable or inert mid-game (see Phase 4 for after game over).

3.7 — Mid-game refresh / reopen by id

Steps

1. Play a few moves online on `/play?id=...`
2. Hard-refresh the page (or leave and reopen the same id from Friends / Active games).
3. Confirm URL stays or returns to `/play?id=...`

Expect

- Position and move list restore from the server.
- WebSocket reconnects; play can continue.
- Clocks rehydrate to sensible values.
- Local SP sessionStorage resume does **not** overwrite the online game.

3.8 — Disconnect messaging (basic)

Steps

1. During an online game with moves, close or kill one browser tab / disconnect network briefly for one player.
2. Observe the other player's status for a few seconds.
3. Reopen / reconnect the disconnected player if practical.

Expect

- Connected player sees a disconnect-related status (e.g. "Opponent disconnected").
- Phase 4 adds the full reconnect countdown + forfeit parity (see 4.5–4.6).
- If the opponent rejoins, a rejoined message / status is acceptable when the server sends it.

3.9 — Classic `/game` fallback

Steps

1. Create or open an online game.
2. Manually navigate one client to `/game?id=<sameId>` (or use a bookmark).
3. Make a move from classic UI; observe the `/play` client (or the reverse).

Expect

- Classic `/game` still loads and can play the same OnlineGame.
- Moves still sync across clients (protocol compatible).
- No server crash.

3.10 — Active games / Friends reopen → `/play`

Steps

1. With an in-progress online game as a participant, open **Active games** list and click the game.
2. From **Friends**, reopen the shared in-progress game if the UI offers it.

Expect

- Play UI users land on `/play?id=...`
- Game resumes as in 3.7.

3.11 — Out-of-time online (if easy)

Steps

1. If game time can be set very short for online invites, let a side flag.
2. Otherwise skip and note "not run".

Expect

- Flagged side loses; both clients show timeout / game over.
- Clocks stop.

3.12 — Single-player regression after OnlineMode

Steps

1. After an online session (or on a fresh `/play`), use **Play Now** vs Brain.
2. Play a move; wait for engine reply; resign once.

Expect

- LocalEngineMode still runs (engine replies).
- No WebSocket errors blocking SP.
- Undo still works in SP; Draw still disabled vs engine.

3.13 — Spectating (see Phase 5)

Steps

1. As a third user, open a live online game from Active games (non-participant), or `/play?id=<id>&mode=watch`.

Expect

- Prefer-Play users land on `/play` in Watch Mode (see §5.1).
- Classic `/watch?id=` remains as fallback (redirects Prefer-Play desktop users to `/play`).

3.14 — Promotion in an online game

Steps

1. Reach a pawn promotion (or set up via a long game / known line).
2. Promote for the human side; confirm opponent sees the promoted piece.

Expect

- Promotion dialog on the promoting player's client.
 - Completed promotion syncs to the opponent.
 - Game continues normally.
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Phase 4 — Draw / rematch / reconnect on /play

Phase 4 completes social online parity on the Play shell: draw offers, rematch with color choice, and disconnect countdown UX (opponent name red + clock hidden; forfeit via existing server reconnect rules).

4.1 — Draw disabled mid-game until legal

Steps

1. Start an online game on /play with two players.
2. Before either side has a legal draw window (e.g. before White has moved, or while it is still your turn), check **Draw**.

Expect

- **Draw** is disabled when offering is not allowed (your turn / no moves yet), matching classic rules.
- After you have moved and it is the opponent's turn, **Draw** becomes enabled.

4.2 — Offer draw / accept

Steps

1. With Draw enabled, click **Draw**.
2. On the opponent client, accept the offer dialog.

Expect

- Offerer sees "Draw offer sent" (or equivalent status).
- Opponent sees a confirm dialog ("Opponent sent a draw offer...").

- After Accept, both clients show a drawn game over; clocks stop.
- Resign / Draw become inactive; **Rematch** becomes available.

4.3 — Offer draw / decline

Steps

1. Offer a draw again in a new game (or if decline mid-game is still open).
2. Opponent clicks **Decline**.

Expect

- Offerer sees decline status.
- Game continues; Draw can be offered again later when legal.

4.4 — Rematch with color choice (accept)

Steps

1. Finish an online game (resign or draw).
2. Click **Rematch** (same rail slot as New game; label is **Rematch** online).
3. In the color dialog, choose **White** (or **Black**).
4. Opponent accepts; confirm the accept dialog mentions the offerer's preferred color and your resulting color.

Expect

- Offerer dialog offers **White / Black / Cancel**.
- After accept, both clients load a new `/play?id=...` game.
- Seats match the choice: offerer gets the color they picked; acceptor gets the other.
- New game clocks / names / board orientation match the new seats.

4.5 — Rematch decline + cancel color dialog

Steps

1. After game over, open Rematch color dialog → **Cancel**.
2. Offer rematch again → opponent **Decline**.

Expect

- Cancel does not send an offer.
- Decline leaves both on the finished game; no new id; Rematch still available to offer again.

4.6 — Disconnect UX (countdown + chrome)

Steps

1. Mid-game with moves played, close one client's tab (or disconnect network) for ~2+ seconds.
2. Watch the remaining client's header and status for up to ~60s.
3. Optionally reconnect the disconnected player before forfeit.

Expect

- After a short grace (~1s), status shows a disconnect countdown.
- Opponent name styling turns red; opponent clock is hidden while disconnected.
- If they rejoin in time, countdown clears and play continues.
- If they do not rejoin, server forfeit / "failed to reconnect" behavior matches classic expectations (game over for the disconnected side).

4.7 — Resign still stable during opponent moves

Steps

1. During an online game, watch **Resign** while the opponent's move animates.

Expect

- Resign does **not** flicker disabled solely because of move animation.
- Resign still ends the game correctly when clicked on your turn / when allowed.

4.8 — Classic /game rematch still works (no color required)

Steps

1. Finish an online game on classic /game.
2. Offer rematch without a color field (legacy path).
3. Accept from the other classic (or /play) client.

Expect

- Rematch creates a new game with the **same** seats as before (legacy default).
- No server validation error on missing offererWantsColor.

4.9 — Single-player regression after Phase 4

Steps

1. On /play, start **Play Now** vs Brain.

2. Confirm Draw stays disabled vs engine; New game still opens the new-game dialog (not rematch color).

Expect

- LocalEngineMode unchanged.
 - No online rematch dialog when not in an online finished game.
-

Phase 5 — Watch / Review (partial)

5.1 — Live watch on /play (slice 1)

Steps

1. Two players start an online game on /play.
2. As a third logged-in user (Prefer-Play), open **Active games** and click that game (or open /watch?id=...).
3. Observe moves as players play; try Resign / Draw / Rematch / dragging pieces.
4. Click Home / leave.

Expect

- Third user lands on /play?id=...&mode=watch (or equivalent after /watch redirect).
- Header shows **Watch Mode**; status indicates watching.
- Board updates when players move; clocks update.
- Resign / Draw / Rematch disabled; pieces cannot be moved.
- Draw / rematch offer dialogs do **not** appear for the watcher (player↔player only).
- Leaving does **not** resign either player.
- Classic /watch still works for clients that do not Prefer-Play (e.g. mobile classic path).

5.2 — Watcher sees disconnect of the correct player

Steps

1. While watching a live game, close one player's tab (e.g. White).
2. Watch status and header chrome for ~60s without that player rejoining.

Expect

- Status like "<WhiteName> disconnected — waiting for rejoin (...countdown...)", **not** "Opponent disconnected".
- The disconnected seat's name is red / clock hidden (the seat that left, not always Black).
- After ~1 minute, that player loses; status like "<WhiteName> failed to reconnect — <BlackName> wins".

- Result matches (e.g. White disconnect → Black wins / 0-1).

Phase 5 remaining (to be filled)

- ☒ Live watch on /play (non-participant)
- ☒ History / PGN review deep-links into /play (Prefer-Play)
- ☐ Optional: update Home/Search links to /play?mode=review directly (redirect covers Prefer-Play)

5.3 — History review on /play

Steps

1. From Home (or Games list), open a finished game row (history).
2. Confirm URL becomes /play?mode=review&id=...&type=history (after /review redirect).

Expect

- Review Mode header; ply nav works (start / back / forward / end / play).
- Moves list matches the finished game and includes the result (1-0 / 0-1 / 1/2-1/2).
- Status shows the outcome (e.g. "White wins", "Black resigned", "Draw").
- Header clocks show the times stored on the current ply (when moves include whiteTimer/blackTimer or moveTime); clocks do not run or flag.
- Pieces are not playable.
- Mobile UA still uses /mobile-review (unchanged).

5.4 — PGN review on /play (if Search available)

Steps

1. From Search, open a PGN game.
2. Confirm /play?mode=review&...&type=pgn.

Expect

- PGN converts and ply nav works from the start position.
- No OnlineMode / WebSocket play session attached.

Phase 6 — Practice / Debug on /play (slice 1)

6.1 — Prefer-Play Debug entry

Steps

1. As Admin or Partner (Prefer-Play), click **Debug** in the top bar (or open `/play?mode=practice`).
2. Confirm URL is `/play?mode=practice`.

Expect

- Header shows **Practice Mode**.
- Status like "Practice — both sides".
- Both White and Black pieces are playable on their turns (self-play).
- Draw is disabled; Rematch/New game stays available for a fresh practice game.
- No engine thinking / no brain HTTP after your moves.
- Members do not see Debug; `/play?mode=practice` as Member shows an error / does not start practice.

6.2 — Undo / resign / exit

Steps

1. Play a few plies as both colors.
2. Undo once (single ply), then redo.
3. Resign (confirm "Are you sure?").
4. Start another practice via New game, then Exit mid-game after a move (confirm "Are you sure?").

Expect

- Undo removes one ply at a time and stays enabled until the start position (all saved plies undone); redo stays enabled while the redo stack has items.
- Position Setup is disabled for the whole practice session.
- Resign ends the game for the side to move.
- Exit after moves confirms with "Are you sure?" then leaves.
- Classic `/game?gameType=3` still works for non-Prefer-Play clients.

Phase 6 remaining

- ☒ PracticeMode on `/play` for Prefer-Play Admin/Partner
- ☐ Optional: Electron Debug entry parity
- ☐ Retire any remaining "debug always classic" comments once QA signed off

Phase 7 — Position Setup / Config as modes (slice 1)

7.1 — Position Setup attaches PositionSetupMode

Steps

1. As Admin/Partner on `/play` (local SP idle or game over / zero moves), open Position Setup.
2. Confirm header shows **Position Setup**.
3. Place pieces, validate, Play from position.
4. Cancel setup (restore) from a local SP game over / zero-move board.

Expect

- Setup dock works as before; resign/draw/undo stay disabled while in setup.
- After Play from position, LocalEngine play starts (engine may move if it is Black's turn).
- Cancel restores the previous board snapshot.

7.2 — Online / Practice / Watch cannot open setup or config

Steps

1. During an online `/play` game (or watch), try Position Setup and Configuration.
2. In Practice / Debug, try Position Setup and Configuration.

Expect

- Buttons stay disabled (or show a short status if somehow triggered).
- Opening setup never leaves OnlineMode attached to a mutated board.
- Members still do not see the advanced docks.

7.3 — Configuration attaches ConfigurationMode

Steps

1. From idle `/play` (Admin/Partner), open Configuration, edit/save or discard, exit.

Expect

- Header shows **Configuration mode**.
- Unsaved discard confirm still works.
- Exiting restores LocalEngine caps when returning to local play.

Phase 7 remaining

- ☒ PositionSetupMode + ConfigurationMode plugins
 - ☒ Online/watch/practice leak guards for setup/config
 - ☐ Optional: Prefer-Play deep-link for setup
 - ☐ Optional: move more setup board logic out of `desktop-play.js`
-

Phase 8 — Mobile session package (slice 1: review)

8.1 — Mobile history review uses ReviewMode

Steps

1. On a phone UA (or narrow window), open Home → a finished game → `/mobile-review?type=history&id=...`
2. Use start / prev / next / end / play on the review toolbar.
3. Open the moves sheet and jump around if available.

Expect

- Page still uses mobile CSS (not desktop `/play`).
- In the browser console, `window.__SHMERLING_MOBILE_REVIEW_SESSION__` is set after load.
- Ply nav works; board and clocks update; pieces are not playable.
- Desktop Prefer-Play still uses `/play?mode=review` (unchanged).

8.2 — Mobile SP uses LocalEngineMode (client brain)

Steps

1. Restart server, hard-refresh. On a phone UA (or narrow window), start a **new** single-player game from mobile Home / Play Now → `/mobile-game`.
2. Play a few moves as White (and optionally start a second game as Black so the engine moves first).
3. In DevTools Network: confirm `/api/brain/compute-move` after your moves (engine thinking).
4. In console: `window.__SHMERLING_MOBILE_LOCAL_ENGINE_SESSION__` should be set.

Expect

- Mobile CSS/DOM unchanged (not desktop `/play`).
- Engine replies; board animates; clocks and moves list update.
- Resign / game over still work.
- Classic desktop `/game` SP (no Prefer-Play) still uses **server** brain (no mobile adapter).

8.3 — Mobile online uses OnlineMode

Steps

1. Restart server, hard-refresh. On a phone UA (or narrow window), start or join an **online** game so you land on `/mobile-game` as a participant (not watch).
2. Confirm both sides can move; clocks update; resign / draw / disconnect messaging still work.

3. In console: `window.__SHMERLING_MOBILE_ONLINE_SESSION__` is set after load.
4. Optionally rematch — page should reload into the new game id.

Expect

- Mobile CSS/DOM unchanged (not desktop `/play`).
- Classic WS is deferred (`[chessboard] classic WS deferred to mobile OnlineMode in console`).
- SP mobile games still use `LocalEngineMode` (`__SHMERLING_MOBILE_LOCAL_ENGINE_SESSION__`), not `OnlineMode`.

8.4 — Mobile watch uses OnlineMode (watcher)

Steps

1. On a phone UA, open a live public game as spectator via Home → Games / active list → `/watch?id=...`
2. Confirm moves update the board and clocks; pieces are not playable.
3. Console: `__SHMERLING_MOBILE_ONLINE_SESSION__` set; classic WS deferred (when mobile shell loads).

Expect

- Session `OnlineMode` watcher path works (read-only live updates).
- Desktop Prefer-Play still uses `/play?id=&mode=watch`.
- Desktop classic `/watch` (non-mobile UA) still `game.ejs`.

Known gap (deferred polish)

Mobile watch may still render the **classic web** `game.ejs` **look** instead of the mobile-game shell styling (narrow desktop / UA detection). Treat full mobile CSS for watch as a follow-up improvement, not a Phase 8 blocker.

Phase 8 remaining

- ☒ Thin `ReviewMode` adapter on `/mobile-review`
 - ☒ Mobile SP `LocalEngineMode` (`clientEngine` + HTTP `EnginePort`)
 - ☒ Mobile `OnlineMode` (participant `/mobile-game`)
 - ☒ Mobile watch `OnlineMode` watcher (session path)
 - ☐ **Deferred:** mobile watch visual shell (true `mobile-game` CSS/layout, not classic web UI)
-

Deferred improvements (post Phase 8)

Item	Notes
Mobile watch visual polish	/watch should consistently use mobile-game shell CSS on phones; today session wiring works but look may still be classic web UI
Phase 9 optional	prefs/themes adapters; remove legacy <code>runEngineMove</code> fallback in <code>desktop-play.js</code>
Mobile board blink (fixed)	Soft <code>patchBoardFromState + animateMove({ skipFinalSync })</code> so moves don't wipe all 64 squares

Mobile board polish — no full-board blink after moves

Steps

1. Restart server; hard-refresh `/mobile-game` (or open a mobile UA game).
2. Play a move as human; wait for engine/opponent reply (with animation).
3. Play several more moves (including a capture if easy).

Expect

- Board does **not** flash / fully refresh after each move.
- Pieces update in place; opponent/engine animation settles without a snap-back blink.
- Clocks, captures, and moves list still update normally.
- Brain replies on every turn (not only the first); no “Something went wrong” after the engine’s first move.
- No full-board “Engine thinking...” flash (errors may still flash).

Phase 9 — Desktop adapters (slice 1: brain EnginePort)

9.1 — Web Play still uses HTTP brain

Steps

1. Restart server, hard-refresh `/play`.
2. Start a single-player game vs brain; play a few moves.
3. Optionally open evaluation overlay if available.

Expect

- Engine moves still arrive via `/api/brain/*` (Network tab).
- No console error about `ShmerlingCreateEnginePort`.
- Session / `LocalEngineMode` behavior unchanged.

9.2 — Electron Play still uses IPC brain (if desktop build available)

Steps

1. Run desktop app Play shell.
2. Start SP game and confirm engine replies.

Expect

- Brain uses IPC (`brain:computeMove`), not web `/api/brain` to a remote host.
- Abort / new game still stops thinking.

Phase 9 remaining

- ☒ Extract `brainHttp / brainIpc / createEnginePort`; slim `desktop-engine.js`
- ☒ Document `EnginePort` on session contracts
- ☐ Optional: prefs/themes adapters beyond existing split
- ☐ Optional: remove legacy shell `runEngineMove` fallback path

Phase 10 — Deprecate classic `/game` (slice 1: soft redirects)

Goal: Prefer-Play desktop users leave classic `/game` play rendering via redirects. Do **not** delete `chessboard.js` yet.

10.1 — Prefer-Play `/game` → `/play`

Steps

1. Restart server, hard-refresh (desktop Prefer-Play / logged-in).
2. Visit `/game` → should land on `/play`.
3. Visit `/game?gameType=1&newGame=1&color=white&engine=brain43&difficulty=3` → `/play?newGame=1&...`
4. Visit `/game?gameType=3` (Admin/Partner) → `/play?mode=practice`.
5. Escape hatch: `/game?classic=1` starts a default classic SP game (then `/game?id=...&classic=1`).
Or use the full URL: `/game?classic=1&gameType=1&newGame=1&color=white&engine=brain43&difficulty=3`.

Expect

- Soft redirects only (302); classic UI still exists behind `?classic=1`.

- Mobile `/game` still → `/mobile-game` (unchanged).
- SP reopen by `/game?id=` (non-online) still classic during this window.

10.2 — Prefer-Play friend join without classic hop

Steps

1. Restart server; hard-refresh both Prefer-Play clients.
2. As White, invite a friend to a game.
3. As Black, accept the invite.
4. Observe Black's navigation URL (and White's after accept).

Expect

- Black lands on `/play?id=...` (not `/game?...&joinGame=`).
- White lands on `/play?id=...`
- Both can move; OnlineMode works.
- Classic non-Prefer-Play (if tested) still uses `/game?joinGame=` after accept.

Phase 10 remaining

- ☒ Soft Prefer-Play redirects `/game` → `/play` (+ `classic=1` escape)
- ☒ Resolve `/play` `joinGame` bounce (Prefer-Play accept → `/play?id=` without classic hop)
- ☒ **Won't do:** migrate SP `/game?id=` reopen to `/play` (rare; Prefer-Play SP is client LocalEngineMode; classic reopen stays on `/game`)
- ☐ After parity sign-off: remove `/game` play render + retire `chessboard.js` play path
- ☐ Short public deprecation notice / final cutover

Prefer-Play UCI (Stockfish) — web only

Ops prerequisite: install Stockfish on the server host and set `STOCKFISH_PATH` to the binary (or ensure `stockfish` is on `PATH`). Restart `npm start` after changing the env. Do **not** expect Stockfish on classic `/game` or Electron for this slice.

UCI.1 — Stockfish appears when available

Steps

1. With Stockfish configured, restart the server and hard-refresh `/play`.
2. Open **New game** / Play Now engine dropdown.

Expect

- **Stockfish** is listed with Brain 4.x options.
- Starting a game as White vs Stockfish: after your move, Stockfish replies within the configured thinking time.
- Moves are legal and animate like Brain games.

UCI.2 — Unavailable when binary missing

Steps

1. Unset `STOCKFISH_PATH` / remove `stockfish` from `PATH` (or point at a bad path).
2. Restart server; hard-refresh `/play`.
3. Open New game engine dropdown.

Expect

- Stockfish is **not** listed (or is unavailable).
- Brains still work; Play shell does not crash.

UCI.3 — Think time only

Steps

1. Preferences → thinking time 2s; play vs Stockfish.
2. Increase thinking time; play again.

Expect

- Search duration roughly tracks think time (no Elo/Skill UI required).

Security remediations smoke (Aug 2026)

Run after restarting the server and a hard-refresh. Automated coverage: `npm run test:web:api` (includes `test/security.http.test.js`).

SEC.1 — Search, history click → review on `/play`

Steps

1. Open Search; run a query that returns rows.
2. Click a result row.
3. From Home/history, click one of your finished games.

Expect

- Search results render; row click navigates (inline `onclick` works).
- Review opens on Prefer-Play `/play?mode=review&...` (desktop web) with clocks/moves.
- No blank page / CSP console errors blocking navigation.

SEC.2 — Start a new Prefer-Play game

Steps

1. From Home, Play Now → start vs Brain on `/play`.
2. Make a move; confirm engine replies.

Expect

- Game starts; WebSocket connects; no HTTPS upgrade failures on `http://localhost`.

SEC.3 — Friend invite has no Allow undo

Steps

1. Open Friends → invite gear / invite options.

Expect

- Time, color, friendly/private options appear.
- **Allow undo** checkbox is absent; invite still sends successfully.

SEC.4 — Private game not listed / not readable by stranger

Steps

1. Start or invite a **private** online/SP game as user A.
2. As user B (logged in, not a participant), open Active games / try opening A's private finished game moves if you have an id.

Expect

- Private game does not appear in public active listings.
- Stranger cannot load private moves/review (forbidden / redirect home).

Sign-off

Field	Value
Tester	

Field	Value
Date	
Build / branch	
Role(s) tested	
Web / Desktop	
Phase 1 result	Pass / Fail
Phase 2 result	Pass / Fail
Phase 3 result	Pass / Fail
Phase 4 result	Pass / Fail
Phase 5 (watch slice) result	Pass / Fail
Phase 6 (practice slice) result	Pass / Fail
Phase 7 (setup/config) result	Pass / Fail
Phase 8 (mobile review) result	Pass / Fail
Phase 8 (mobile SP LocalEngine) result	Pass / Fail
Phase 8 (mobile OnlineMode) result	Pass / Fail
Phase 8 (mobile watch) result	Pass / Fail
Phase 9 (brain adapters) result	Pass / Fail
Phase 10 (deprecation redirects) result	Pass / Fail
Security smoke (SEC.1–SEC.4)	Pass / Fail
Notes	